

Case Study 3

AKSTA Statistical Computing

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AI usage	Task a	Task b	Task c	Task d	Task e
<i>Correcting faulty code</i>	medium	medium	medium	medium	medium
<i>Improving existing code</i>	no	no	no	no	no
<i>Generating new code</i>	no	no	no	no	no
<i>Layout and formatting</i>	no	no	no	no	no
<i>Text correction and rewording</i>	heavy	heavy	heavy	heavy	heavy

AI usage	Task f	Task g	Task h	Task i	Task j
<i>Correcting faulty code</i>	medium	medium	medium	medium	medium
<i>Improving existing code</i>	no	no	no	no	no
<i>Generating new code</i>	no	no	no	no	no
<i>Layout and formatting</i>	no	no	no	no	no
<i>Text correction and rewording</i>	heavy	heavy	heavy	heavy	heavy

AI usage	Task k	Task l	Task m	Task n	Task o
<i>Correcting faulty code</i>	medium	medium	medium	medium	medium
<i>Improving existing code</i>	no	no	no	no	no
<i>Generating new code</i>	medium	no	no	no	yes
<i>Layout and formatting</i>	no	no	no	no	no
<i>Text correction and rewording</i>	heavy	heavy	no	heavy	heavy

Table 1: AI usage for this solution.

Data

Load the data set you exported in the final Task of Case Study 2. Eliminate all observations with missing values in the income status variable.

As a reminder, the data set includes world data from 2020, focusing on:

- **Education Expenditure (% of GDP)**
- **Youth Unemployment Rate (15-24 years)**
- **Net Migration Rate** (difference between the number of people entering and leaving a country per 1,000 persons)

for most world entities in 2020. The data was downloaded from <https://www.cia.gov/the-world-factbook/about/archives/>. Additional information on continent, subcontinent/region and income status was appended to the dataset in Case Study 2.

Setup This setup includes loading the necessary libraries and reading the data from the provided `file_out2.csv`. We drop the unnecessary index column, filter for rows that have no NA in `status` and

convert status, low_yu and high_nmr to factors.

```
# load libraries
```

```
library(readr)
```

```
library(dplyr)
```

```
##
```

```
## Attache Paket: 'dplyr'
```

```
## Die folgenden Objekte sind maskiert von 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## Die folgenden Objekte sind maskiert von 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
library(stringr)
```

```
library(readxl)
```

```
library(tidyr)
```

```
library(ggplot2)
```

```
# read data
```

```
data = read_csv("file_out2.csv", na=c("", ".")) %>%
```

```
  select(`...1`) %>%
```

```
  filter(!is.na(status)) %>%
```

```
  mutate(status = factor(status,
```

```
    levels = c("H", "UM", "LM", "L"),
```

```
    ordered = TRUE),
```

```
    low_yu = factor(low_yu,
```

```
    levels = c("yes", "no")),
```

```
    high_nmr = factor(high_nmr,
```

```
    levels = c("yes", "no")),
```

```
  )
```

```
## Registered S3 method overwritten by 'bit64':
```

```
## method from
```

```
## print.bitstring tools
```

```
## New names:
```

```
## Rows: 227 Columns: 11
```

```
## -- Column specification
```

```
## ----- Delimiter: "," chr
```

```
## (7): country, ISO3, continent, subcontinent, status, low_yu, high_nmr dbl (4):
```

```
## ...1, expenditure, youth_unempl_rate, net_migr_rate
```

```
## i Use `spec()` to retrieve the full column specification for this data. i
```

```
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
## * `` -> `...1`
```

```
# show structure
```

```
str(data)
```

```
## tibble [216 x 10] (S3: tbl_df/tbl/data.frame)
```

```
## $ country      : chr [1:216] "Afghanistan" "Albania" "Algeria" "American Samoa" ...
```

```
## $ ISO3         : chr [1:216] "AFG" "ALB" "DZA" "ASM" ...
```

```
## $ continent    : chr [1:216] "Asia" "Europe" "Africa" "Oceania" ...
```

```
## $ subcontinent : chr [1:216] "Southern Asia" "Southern Europe" "Northern Africa" "Polynesia" ..
```

```
## $ status       : Ord.factor w/ 4 levels "H"<"UM"<"LM"<...: 4 2 3 2 1 3 1 2 2 1 ...
```

```
## $ expenditure      : num [1:216] 4.1 3.6 NA NA 3.2 3.4 NA 5.5 2.7 5.5 ...
## $ youth_unempl_rate: num [1:216] 17.6 31.9 39.3 NA NA 39.4 NA 23.7 36.3 NA ...
## $ net_migr_rate    : num [1:216] -0.1 -3.3 -0.9 -26.1 0 -0.2 2.1 -0.1 -5.5 8.4 ...
## $ low_yu           : Factor w/ 2 levels "yes","no": 2 2 2 NA NA 2 NA 2 2 NA ...
## $ high_nmr         : Factor w/ 2 levels "yes","no": 2 2 2 2 2 2 1 2 2 1 ...
```

Tasks:

a. Education expenditure in different income levels

Using **ggplot2**, create a density plot of the education expenditure grouped by income status. The densities for the different groups are superimposed in the same plot rather than in different plots. Ensure that you order the levels of the income status such that in the plots the legend is ordered from High (H) to Low (L).

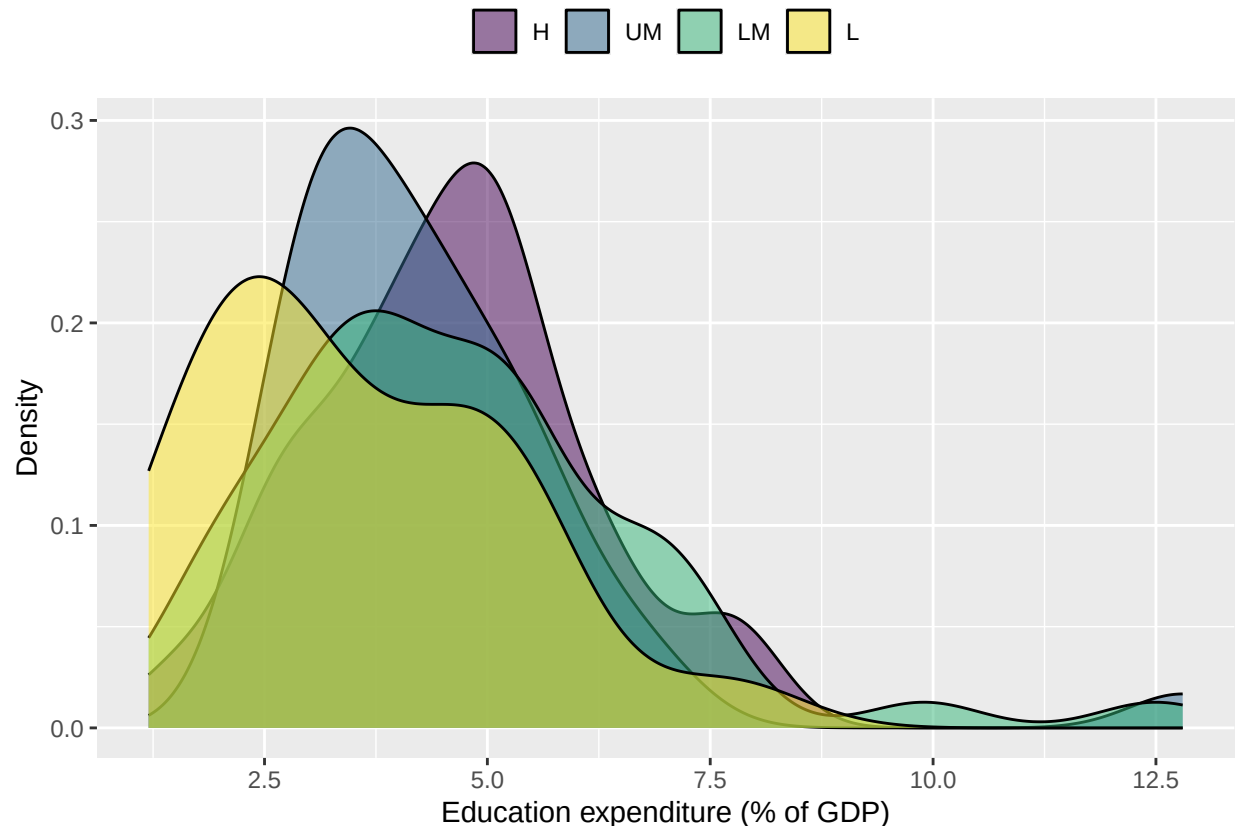
- The color of the density lines is black.
- The area under the density curve should be colored differently among the income status levels.
- For the colors, choose a transparency level of 0.5 for better visibility.
- Position the legend at the top center of the plot and give it no title (hint: use `element_blank()`).
- Rename the x axis as “Education expenditure (% of GDP)”

Comment briefly on the plot.

Answer

```
ggplot(data, aes(x = expenditure, fill = status)) +
  geom_density(colour = "black", alpha = 0.5) +
  labs(x = "Education expenditure (% of GDP)", y = "Density") +
  theme(legend.position = "top",
        legend.title = element_blank(),
        legend.justification = "center")
```

```
## Warning: Removed 48 rows containing non-finite outside the scale range
## (`stat_density()`).
```



The distributions overlap quite strongly, so the differences between income groups are not clearly separated. High-income (H) and upper-middle-income (UM) countries appear to have many observations around roughly 4-5% of GDP and seem somewhat more concentrated than lower-income (L) and lower-middle-income (LM) countries. Lower-income countries are more concentrated at slightly lower education expenditure values, around 2-4% of GDP, but compared to H and UM countries their distribution appears more spread out. The long right tail, especially for the lower-middle-income group, suggests that a few countries spend a much higher share of GDP on education than most others.

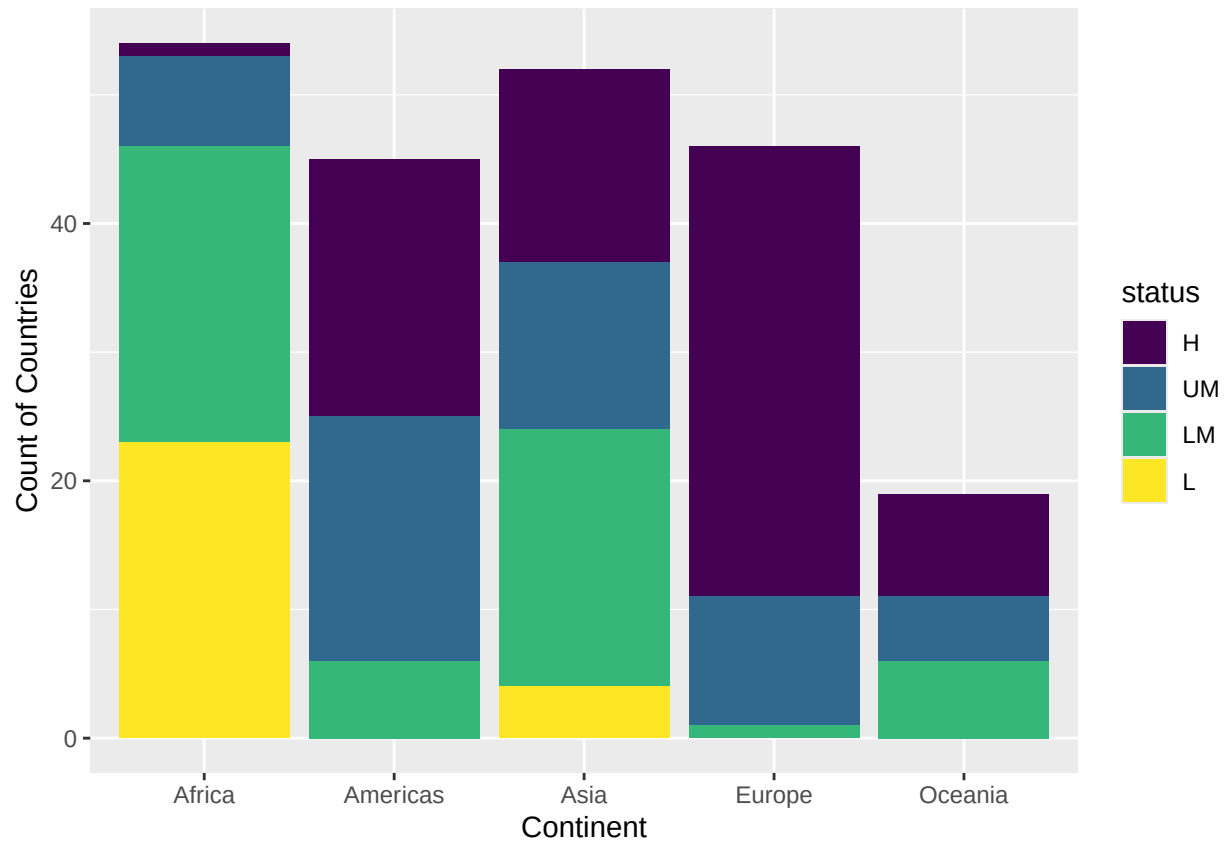
b. Income status in different continents

Investigate how the income status is distributed in the different continents.

- Using **ggplot2**, create a stacked barplot of absolute frequencies showing how the entities are split into continents and income status. Comment the plot.
- Create another stacked barplot of relative frequencies (height of the bars should be one). Comment the plot.
- Create a mosaic plot of continents and income status using base R functions.
- Briefly comment on the differences between the three plots generated to investigate the income distribution among the different continents.

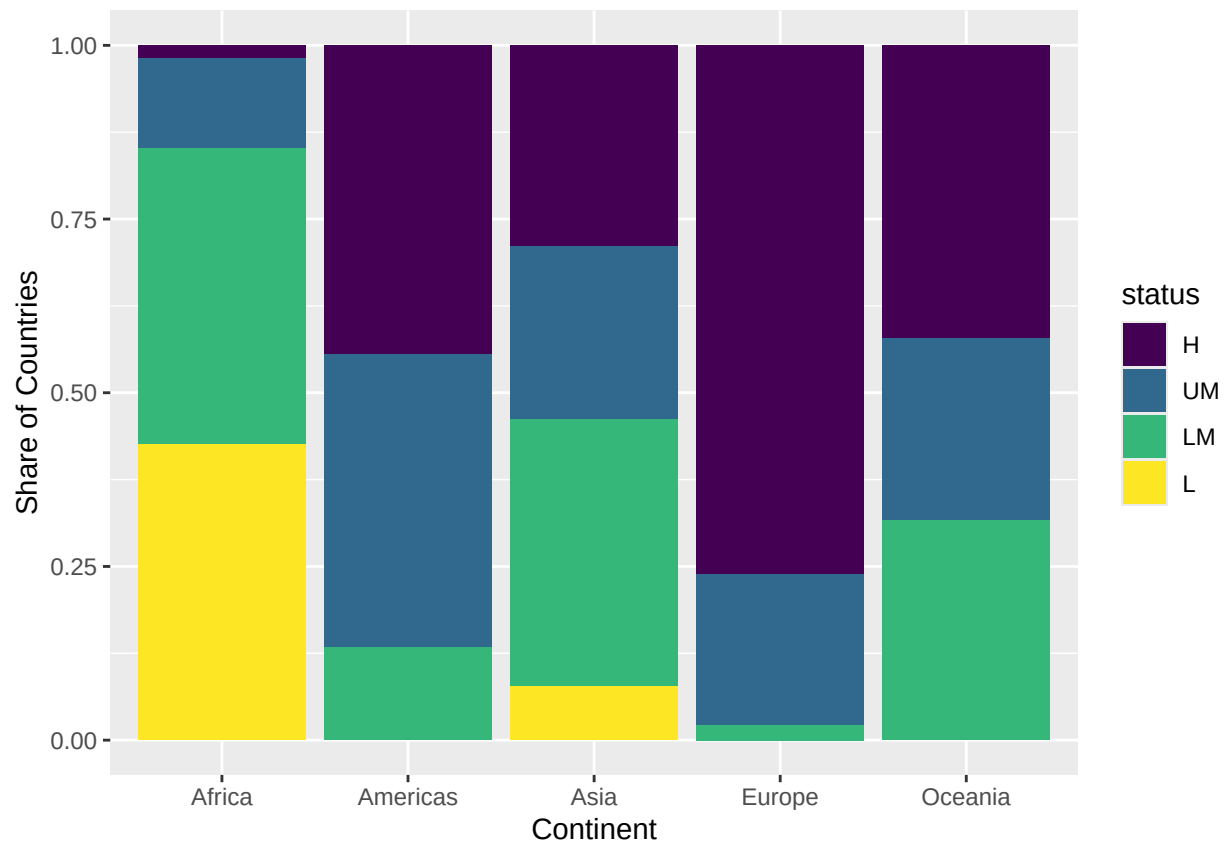
Answer

```
ggplot(data, aes(x = continent, fill = status)) +
  geom_bar(position="stack") +
  labs(x = "Continent", y = "Count of Countries")
```



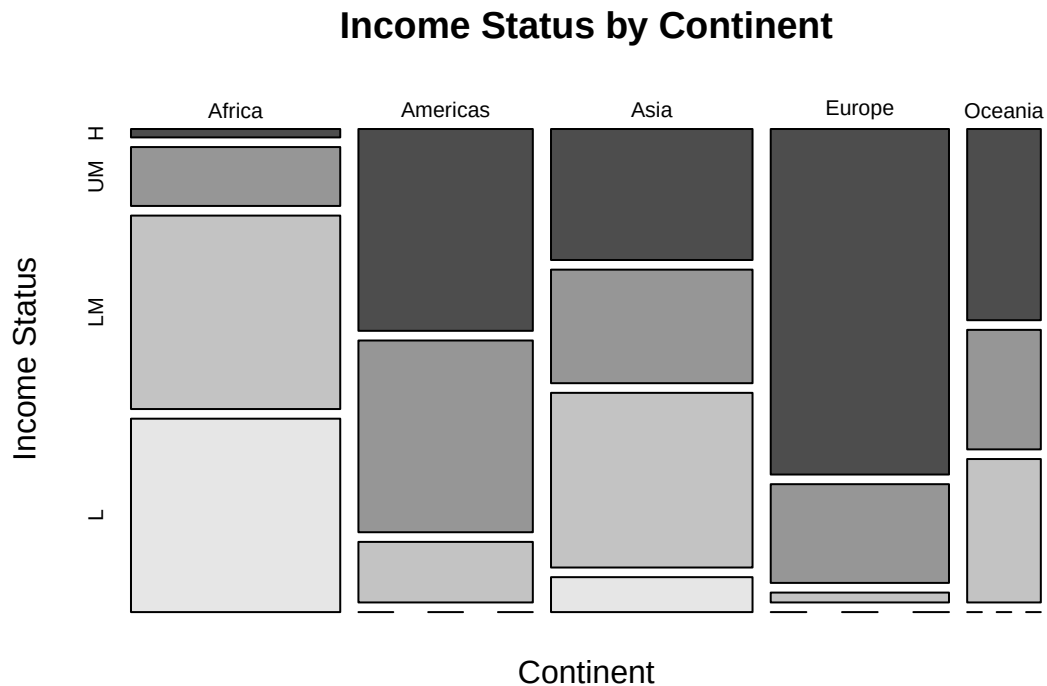
The absolute frequency barplot shows that Africa and Asia contain the largest number of countries in the data set, while Oceania has by far the fewest. In Africa most countries fall into the lower-middle (LM) and low (L) income categories, whereas Europe is dominated by high-income (H) countries. The Americas and Asia show a pretty balanced mix across income groups.

```
ggplot(data, aes(x = continent, fill = status)) +  
  geom_bar(position="fill") +  
  labs(x = "Continent", y = "Share of Countries")
```



Plotting relative frequencies makes the differences in income arrangements more visible and comparable by removing the effect of how many countries each continent contributes. Europe is almost entirely high-income, Oceania and America have a large share of high-income countries as well, while Africa is dominated by low and lower-middle income countries with only a small high-income share. Asia appears comparatively heterogeneous, covering all four income statuses.

```
mosaicplot(table(data$continent, data$status),
  main = "Income Status by Continent",
  xlab = "Continent",
  ylab = "Income Status",
  color = TRUE)
```



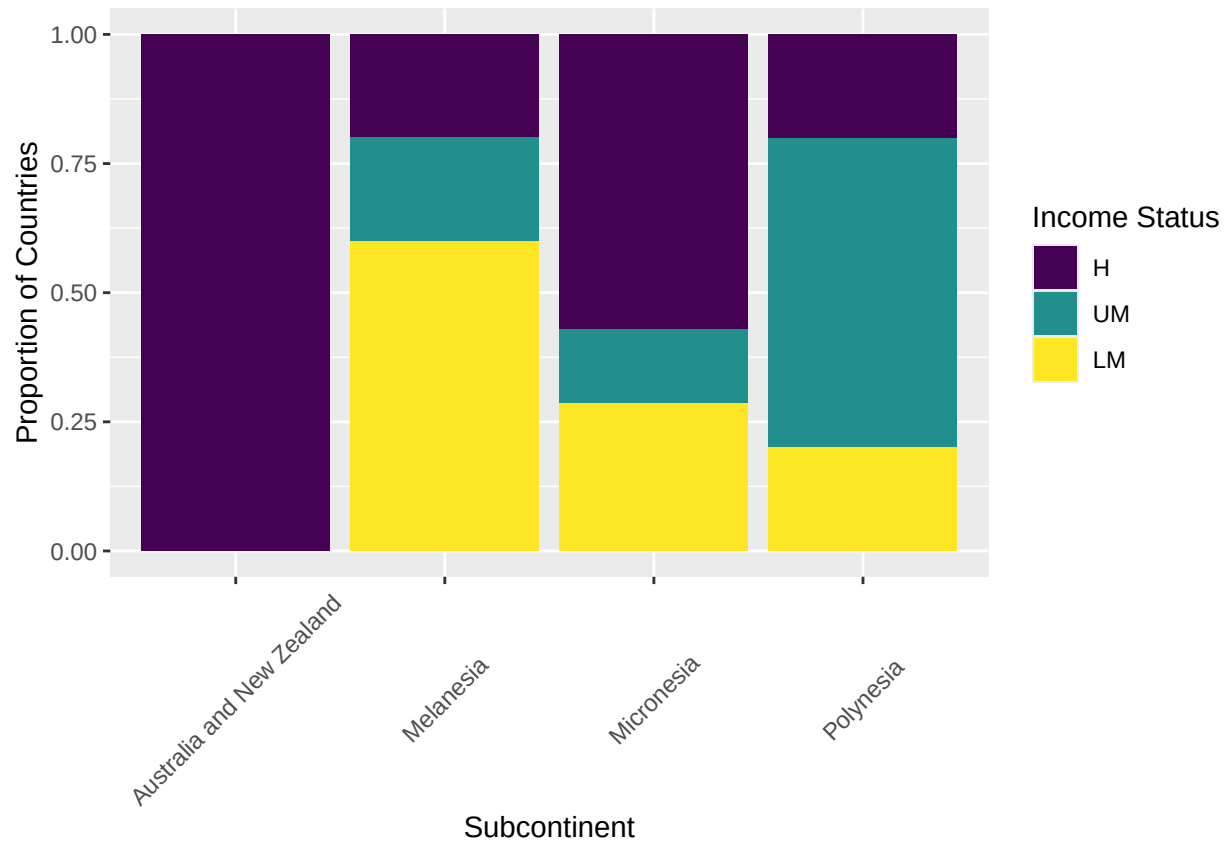
The mosaic plot encodes both pieces of information from the previous plots simultaneously: the width of each column reflects how many countries belong to a continent (like the absolute barplot) and the height of each tile reflects the income composition within that continent (like the relative barplot). The absolute barplot is best for reading off counts but hides composition differences for small continents. The relative barplot makes arrangements comparable across continents but loses the size information. The mosaic plot combines both at the cost of being slightly harder to read precise values from.

c. Income status in different subcontinents

For Oceania, investigate further how the income status distribution is in the different subcontinents. Use one of the plots in b. for this purpose. Comment on the results.

Answer

```
data %>%
  filter(continent == "Oceania") %>%
  ggplot(aes(x = subcontinent, fill = status)) +
  geom_bar(position = "fill") +
  labs(x = "Subcontinent",
       y = "Proportion of Countries",
       fill = "Income Status") +
  theme(axis.text.x = element_text(angle = 45, vjust = 0.5, hjust = 0.5))
```



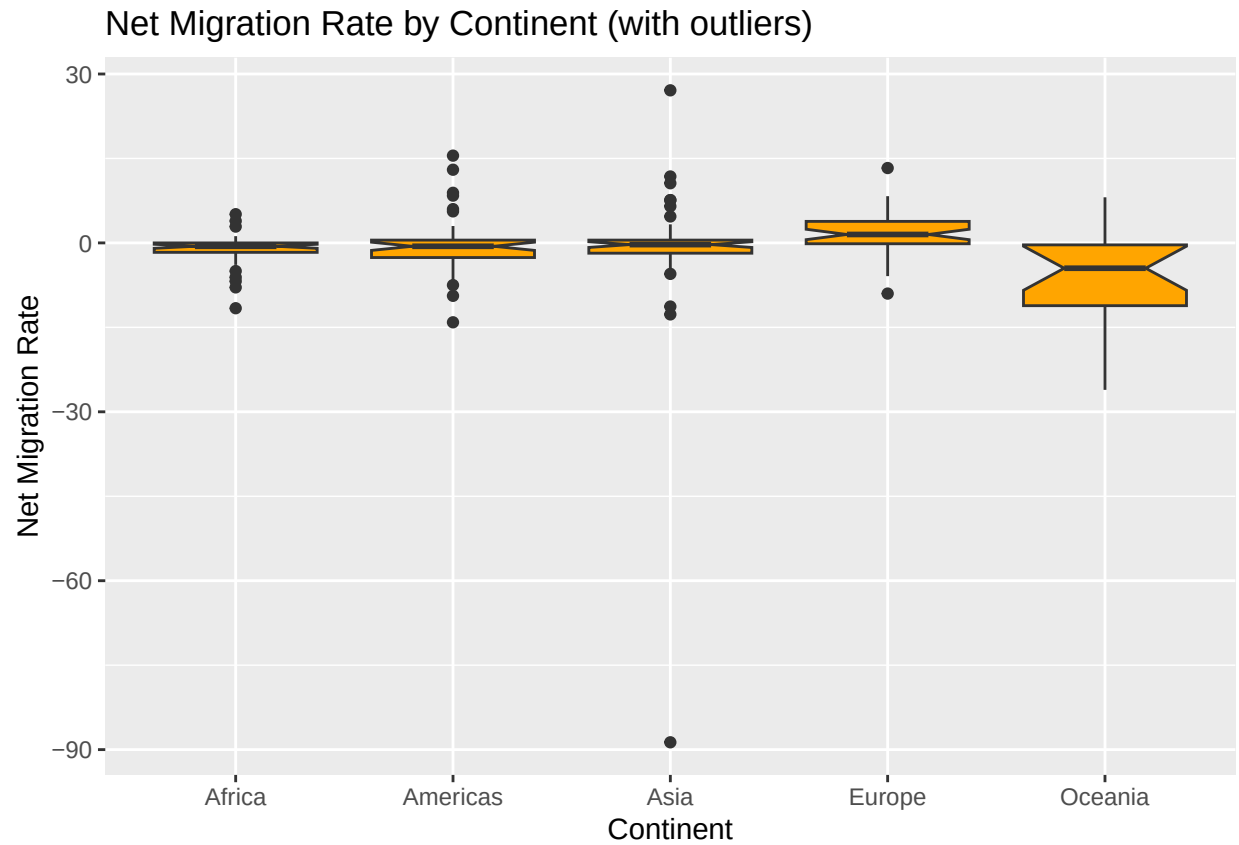
Within Oceania the income status distribution differs strongly between subregions. Australia and New Zealand consist exclusively of high-income countries, while Melanesia and Polynesia are dominated by lower-middle and upper-middle income countries with essentially a moderate amount of high-income entries. It is also worth noting that there are no low-income countries.

d. Net migration in different continents

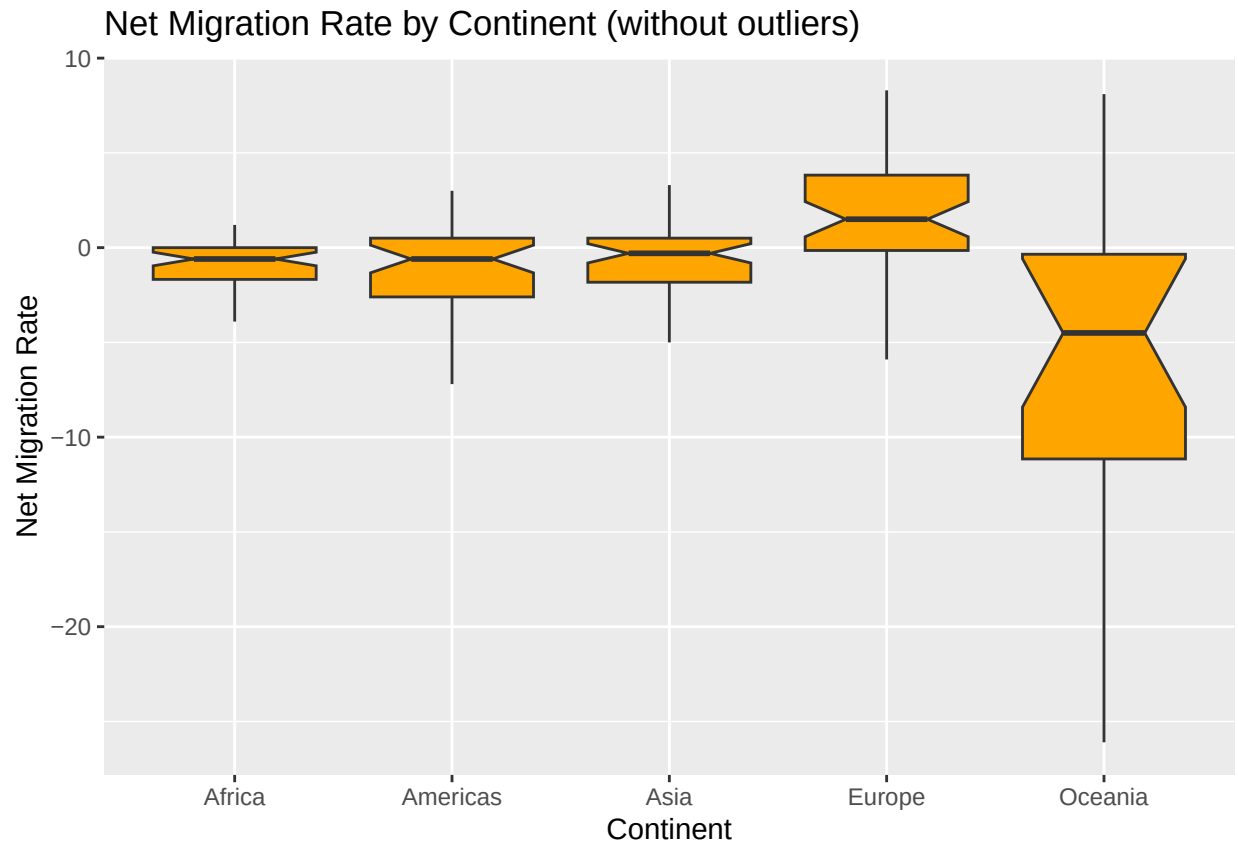
- Using **ggplot2**, create parallel boxplots showing the distribution of the net migration rate in the different continents.
- Prettify the plot (change y-, x-axis labels, etc).
- Identify which country in Asia constitutes the largest negative outlier and which country in Asia constitutes the largest positive outlier.
- Comment on the plot.

Answer

```
# with outliers
ggplot(data, aes(x = continent, y = net_migr_rate)) +
  geom_boxplot(fill = "orange", outliers = TRUE, notch = TRUE) +
  labs(title = "Net Migration Rate by Continent (with outliers)",
       x = "Continent", y = "Net Migration Rate")
```

```
# without outliers  
ggplot(data, aes(x = continent, y = net_migr_rate)) +  
  geom_boxplot(fill = "orange", outliers = FALSE, notch = TRUE) +  
  labs(title = "Net Migration Rate by Continent (without outliers)",  
        x = "Continent", y = "Net Migration Rate")
```



```
# identify largest negative and positive outliers in Asia
data %>%
  filter(continent == "Asia") %>%
  filter(net_migr_rate == min(net_migr_rate, na.rm = TRUE)
         | net_migr_rate == max(net_migr_rate, na.rm = TRUE)) %>%
  select(country, net_migr_rate)
```

```
## # A tibble: 2 x 2
##   country net_migr_rate
##   <chr>      <dbl>
## 1 Lebanon    -88.7
## 2 Syria      27.1
```

Most continents have median net migration rates close to zero or slightly negative, suggesting roughly balanced migration or slight net emigration. Europe stands out with a clearly positive median, while Oceania shows the largest spread, ranging from strong net emigration to net immigration. Overall, the plot suggests that continent-level summaries hide substantial differences between individual countries.

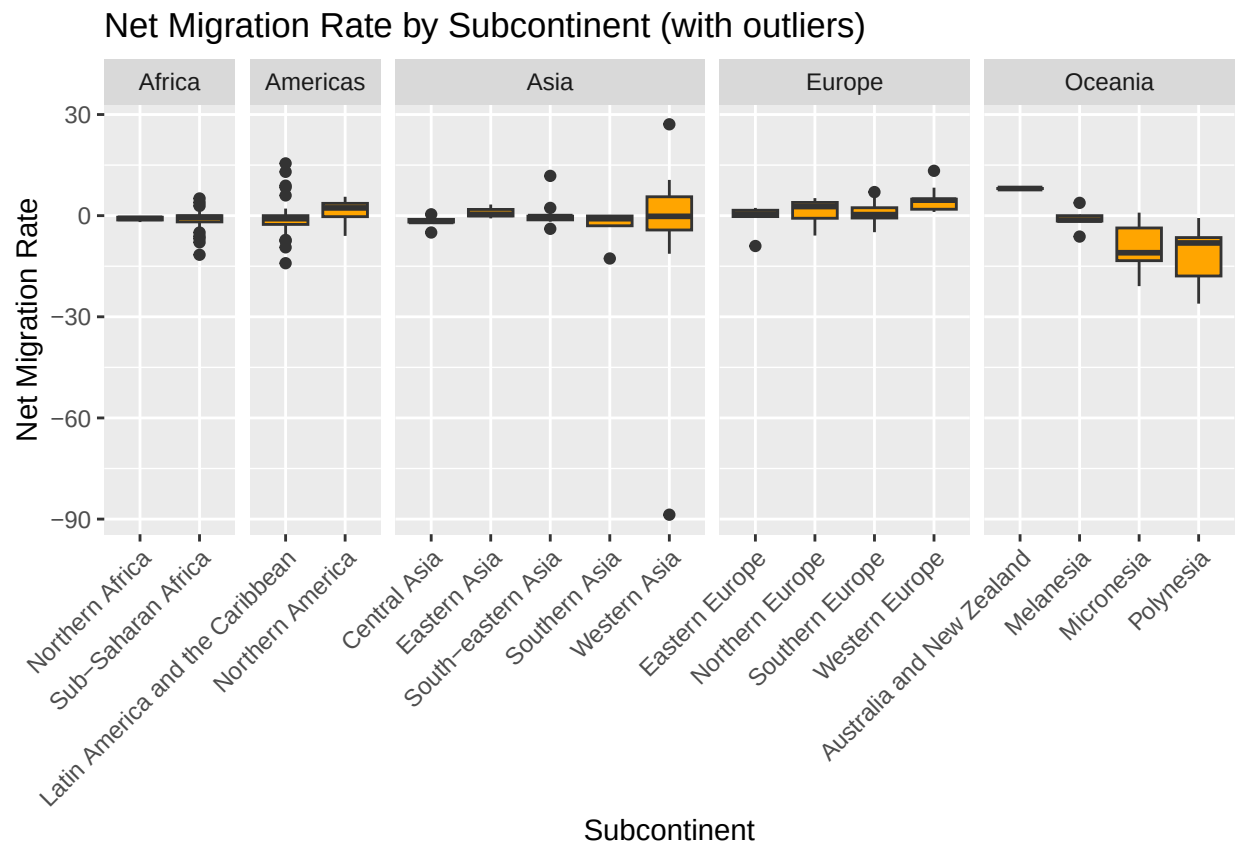
e. Net migration in different subcontinents

The graph in d. clearly does not convey the whole picture. It would be interesting also to look at the subcontinents, as it is likely that a lot of migration flows happen within the continent.

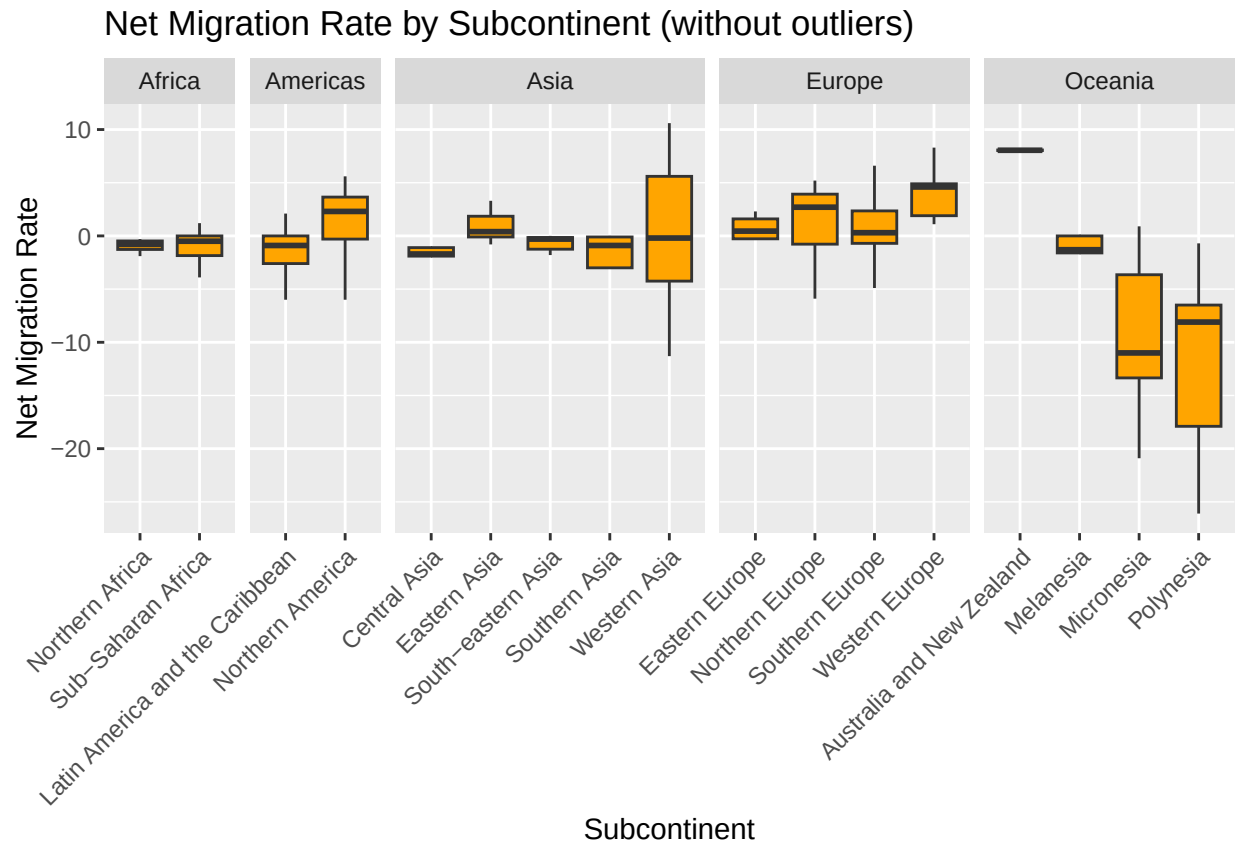
- Investigate the net migration in different subcontinents using again parallel boxplots. Group the boxplots by continent (hint: use `facet_grid` with `scales = "free_x"`).
- Remember to prettify the plot (rotate axis labels if needed).
- Describe what you see.

Answer

```
# with outliers
ggplot(data, aes(x = subcontinent, y = net_migr_rate)) +
  geom_boxplot(fill = "orange", outliers = TRUE) +
  facet_grid(. ~ continent, scales = "free_x", space = "free_x") +
  labs(title = "Net Migration Rate by Subcontinent (with outliers)",
       x = "Subcontinent", y = "Net Migration Rate") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# without outliers
ggplot(data, aes(x = subcontinent, y = net_migr_rate)) +
  geom_boxplot(fill = "orange", outliers = FALSE) +
  facet_grid(. ~ continent, scales = "free_x", space = "free_x") +
  labs(title = "Net Migration Rate by Subcontinent (without outliers)",
       x = "Subcontinent", y = "Net Migration Rate") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Splitting by subcontinent reveals patterns that were hidden in the continent-level boxplots. In Asia, Western Asia shows the widest spread, ranging from negative to strongly positive net migration rates. In the Americas, Northern America is clearly positive, while Latin America and the Caribbean tends to be negative. African subregions are mostly slightly below zero, suggesting moderate net emigration. European subregions are generally positive or close to zero, especially Northern and Western Europe. Oceania shows strong variation, with Australia and New Zealand positive, while Micronesia and Polynesia are strongly negative.

f. Median net migration rate per subcontinent.

The plot in task e. shows the distribution of the net migration rate for each subcontinent. Here you will work on visualizing only one summary statistic, namely the median.

For each subcontinent, calculate the median net migration rate. Then create a plot which contains the sub-regions on the y-axis and the median net migration rate on the x-axis.

- As geoms use points.
- Color the points by continent – use a colorblind friendly palette (see e.g., [here](#)).
- Rename the axes.
- Using `fct_reorder` from the **forcats** package, arrange the levels of subcontinent such that in the plot the lowest (bottom) subcontinent contains the lowest median net migration rate and the upper most region contains the highest median net migration rate.
- Comment on the plot. E.g., what are the regions with the most influx? What are the regions with the most outflux?

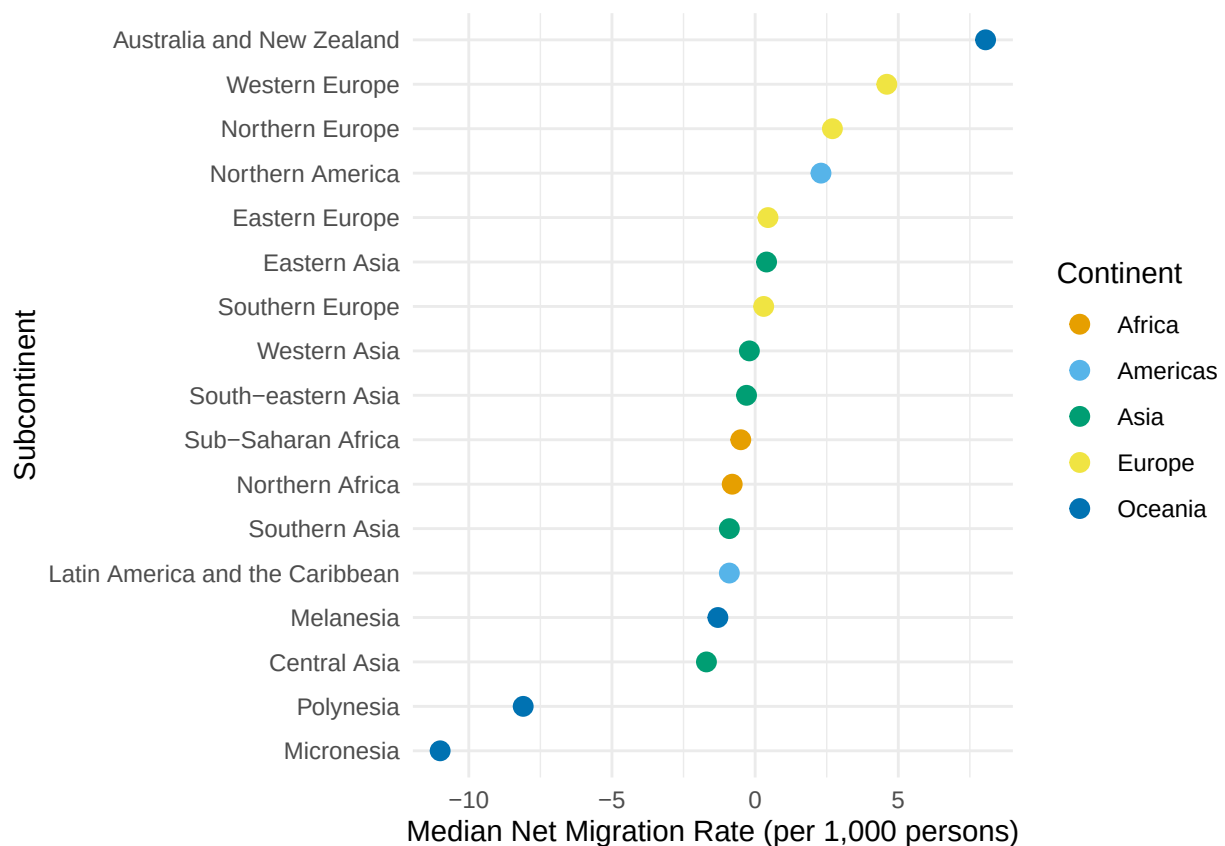
Answer

```
library(forcats)

# colorblind friendly palette
cbPalette <- c("#E69F00", "#56B4E9", "#009E73", "#F0E442",
               "#0072B2", "#D55E00", "#CC79A7", "#000000")

# calculate median net migration rate per subcontinent
subcontinent_migr <- data %>%
  group_by(continent, subcontinent) %>%
  summarize(median_nmr = median(net_migr_rate, na.rm = TRUE), .groups = "drop")

# create the plot
ggplot(subcontinent_migr, aes(x = median_nmr, y = fct_reorder(subcontinent, median_nmr),
  color = continent)) +
  geom_point(size = 3) +
  scale_color_manual(values = cbPalette) +
  labs(x = "Median Net Migration Rate (per 1,000 persons)",
       y = "Subcontinent",
       color = "Continent") +
  theme_minimal()
```



The dot plot shows the median net migration rates for each subcontinent, ordered from the lowest (most outflux) to the highest (most influx). We can see that **Micronesia** and **Polynesia** have the most significant negative median migration rates, indicating substantial outflux. Conversely, **Australia and New Zealand**, **Northern America** and **Western Europe** show the highest positive medians, representing the regions with the most influx. Most other regions cluster around zero, reflecting more balanced migration patterns.

Using different colors for continents helps visualize how migration trends often follow geographic groupings.

g. Median youth unemployment rate per subcontinent

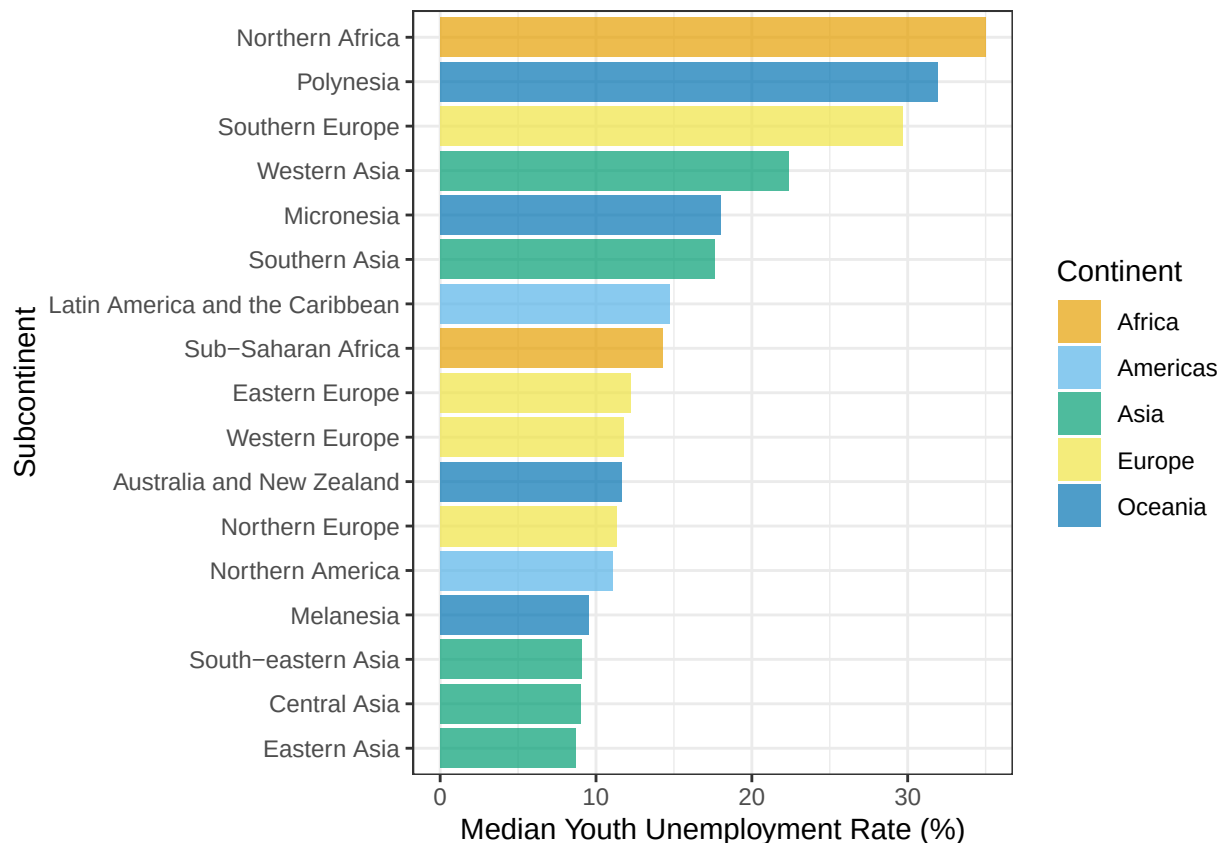
For each subcontinent, calculate the median youth unemployment rate. Then create a plot which contains the sub-regions on the y-axis and the median unemployment rate on the x-axis.

- Use a black and white theme (`?theme_bw()`)
- As geoms use bars. (hint: pay attention to the statistical transformation taking place in `geom_bar()` – look into argument `stat="identity"`)
- Color the bars by continent – use a colorblind friendly palette.
- Make the bars transparent (use `alpha = 0.7`).
- Rename the axes.
- Using `fct_reorder` from the **forcats** package, arrange the levels of subcontinent such that in the plot the lowest (bottom) subcontinent contains the lowest median youth unemployment rate and the upper most region contains the highest median youth unemployment rate.
- Comment on the plot. E.g., what are the regions with the highest vs lowest youth unemployment rate?

Answer

```
# calculate median youth unemployment rate per subcontinent
subcontinent_unempl <- data %>%
  group_by(continent, subcontinent) %>%
  summarize(median_yu = median(youth_unempl_rate, na.rm = TRUE), .groups = "drop")

# create the bar plot
ggplot(subcontinent_unempl, aes(x = median_yu, y = fct_reorder(subcontinent, median_yu),
  fill = continent)) +
  geom_bar(stat = "identity", alpha = 0.7) +
  scale_fill_manual(values = cbPalette) +
  theme_bw() +
  labs(x = "Median Youth Unemployment Rate (%)",
       y = "Subcontinent",
       fill = "Continent")
```



The bar plot reveals substantial differences in youth unemployment across subcontinents. **Northern Africa** and **Southern Europe** stand out with the highest median youth unemployment rates, both exceeding 25-30%. On the other end of the spectrum, **Eastern Asia** and **Central Asia** show the lowest median rates. It is interesting to note that even within continents like Europe, there is a wide gap between the youth unemployment levels in Southern Europe compared to Northern or Western Europe.

h. Median youth unemployment rate per subcontinent – with error bars

The value displayed in the barplot in g. is the result of an aggregation, so it might be useful to also plot error bars, to have a general idea on how precise the median unemployment is. This can be achieved by plotting the error bars which reflect the standard deviation or the interquartile range of the variable in each of the subcontinents.

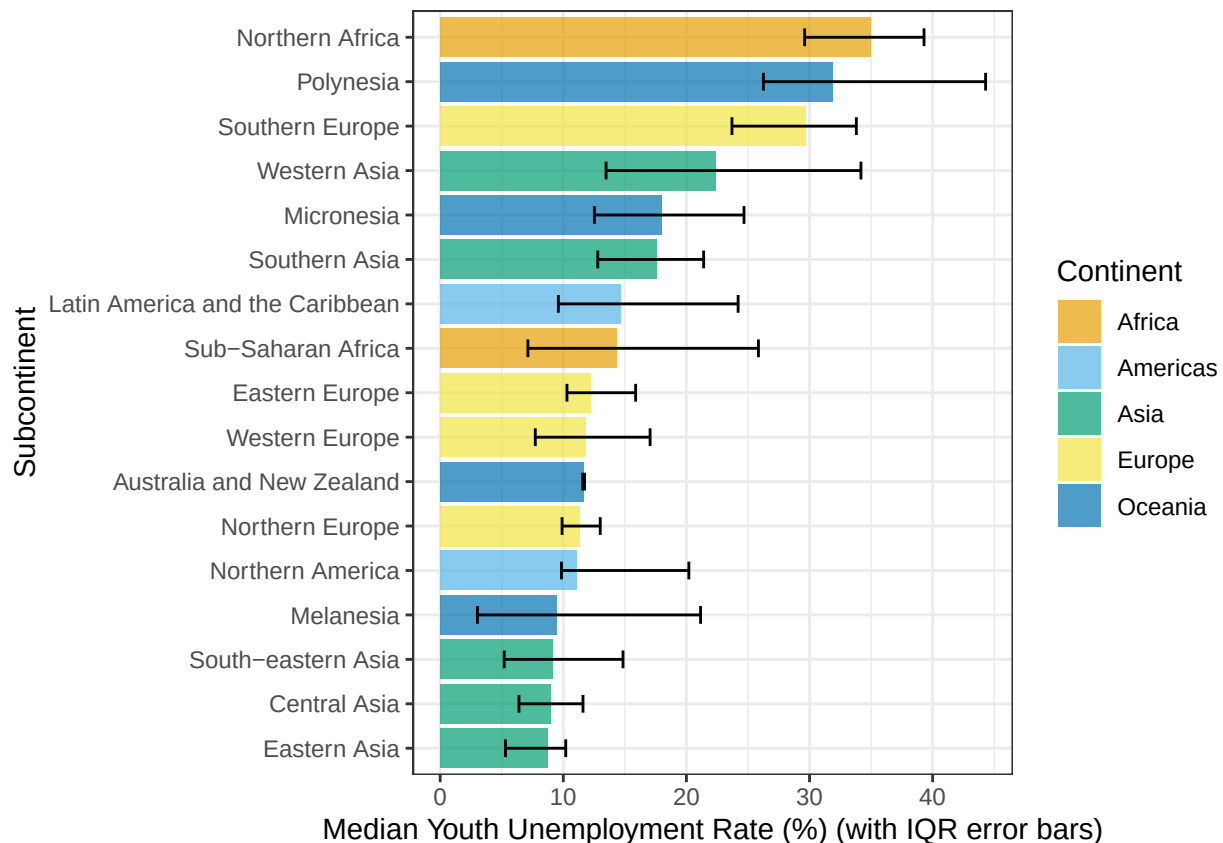
Repeat the plot from Task g. but include also error bars which reflect the 25% and 75% quantiles. You can use `geom_errorbar` in **ggplot2**.

Answer

```
# calculate median and quantiles
subcontinent_unempl_err <- data %>%
  group_by(continent, subcontinent) %>%
  summarize(median_yu = median(youth_unempl_rate, na.rm = TRUE),
            q25 = quantile(youth_unempl_rate, 0.25, na.rm = TRUE),
            q75 = quantile(youth_unempl_rate, 0.75, na.rm = TRUE),
            .groups = "drop")

# create the bar plot with error bars
ggplot(subcontinent_unempl_err, aes(x = median_yu, y = fct_reorder(subcontinent, median_yu),
```

```
fill = continent)) +
  geom_bar(stat = "identity", alpha = 0.7) +
  geom_errorbar(aes(xmin = q25, xmax = q75), width = 0.4, color = "black") +
  scale_fill_manual(values = cbPalette) +
  theme_bw() +
  labs(x = "Median Youth Unemployment Rate (%) (with IQR error bars)",
       y = "Subcontinent",
       fill = "Continent")
```



Adding error bars representing the interquartile range (IQR) provides a clearer picture of the data's variability. While some regions like **Western Asia** have a high median and a relatively wide IQR, others like **South-eastern Asia** show much lower variability. This highlights that in certain regions, youth unemployment is consistently high or low, whereas in others, there is a broader range of experiences across different countries.

i. Relationship between education expenditure and net migration rate

Using **ggplot2**, create a plot showing the relationship between education expenditure and net migration rate.

- Color the geoms based on the income status.
- Add a regression line for each development status (using `geom_smooth()`).

Comment on the plot. Do you see any relationship between the two variables? Do you see any difference among the income levels?

Answer

```
ggplot(data, aes(x = expenditure, y = net_migr_rate, color = status)) +
  geom_point(alpha = 0.6) +
```



```
geom_smooth(method = "lm", se = FALSE) +
labs(x = "Education Expenditure (% of GDP)",
     y = "Net Migration Rate (per 1,000 persons)",
     color = "Income Status") +
theme_minimal()
```

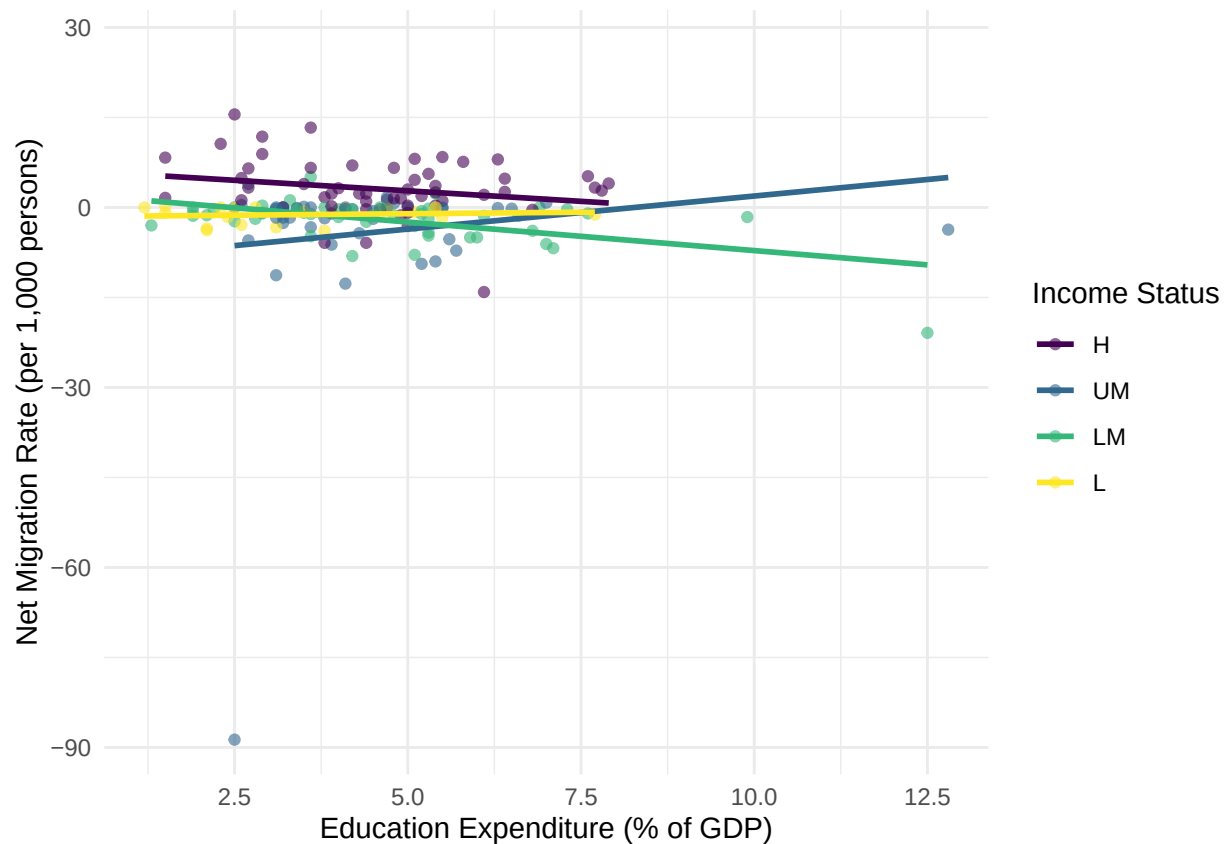
```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 48 rows containing non-finite outside the scale range
```

```
## (`stat_smooth()`).
```

```
## Warning: Removed 48 rows containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```



The scatterplot displays the relationship between education expenditure and net migration rate, categorized by income status. Overall, there is no strong, universal correlation between the two variables across all groups. However, the regression lines suggest some differences: for high-income (H) countries, the relationship appears slightly positive, while for lower-income groups, the lines are flatter or even slightly negative. This might indicate that in wealthier nations, higher investment in education coincides with higher net immigration, potentially due to broader economic opportunities that attract migrants.

j. Relationship between youth unemployment and net migration rate

Create a plot as in Task i. but for youth unemployment and net migration rate. Comment briefly.

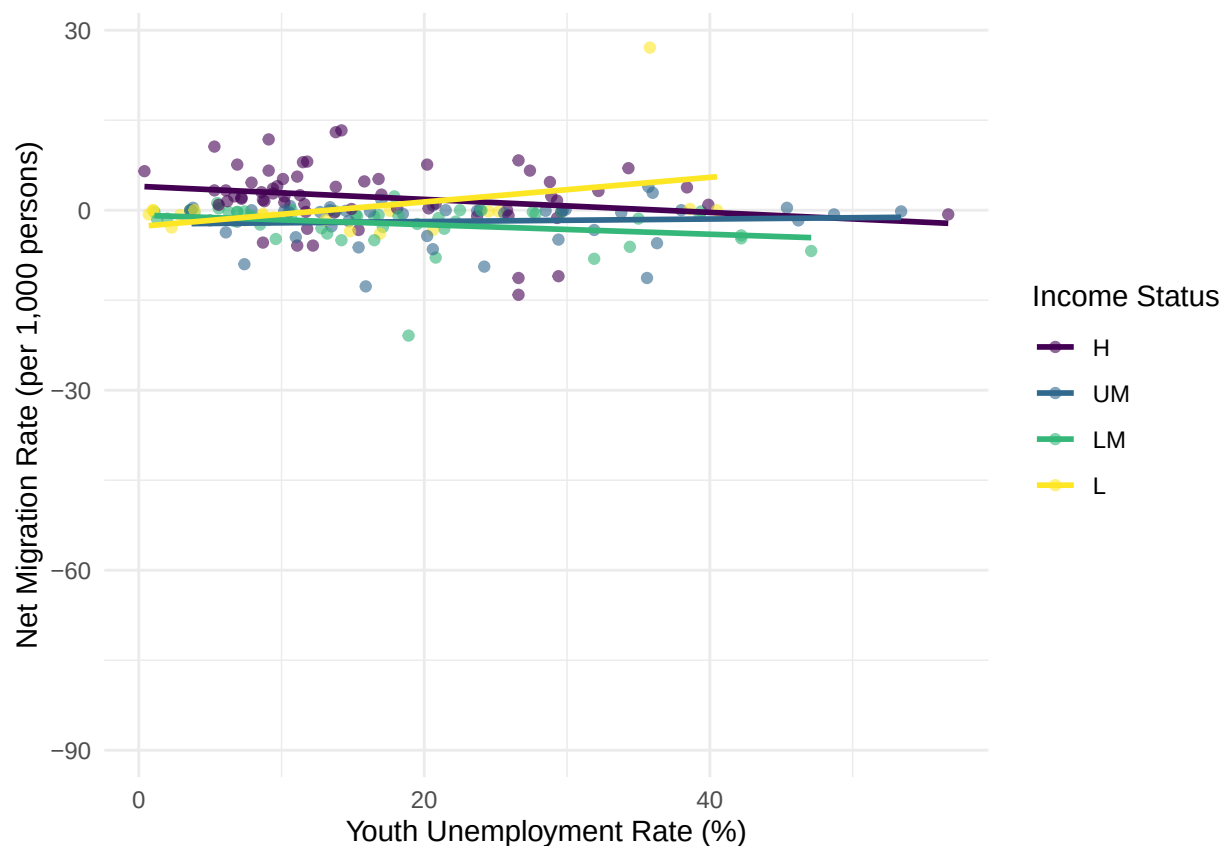
Answer

```
ggplot(data, aes(x = youth_unempl_rate, y = net_migr_rate, color = status)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x = "Youth Unemployment Rate (%)",
       y = "Net Migration Rate (per 1,000 persons)",
       color = "Income Status") +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 37 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 37 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



This plot explores the relationship between youth unemployment and net migration. Generally, there is a weak negative relationship for many groups, suggesting that regions with higher youth unemployment may experience slightly higher outflux (lower net migration). This pattern is most visible in the regression lines for lower-middle (LM) and upper-middle (UM) income status countries. For high-income countries, the relationship remains relatively flat, perhaps because migration to these countries is driven by factors other than the immediate youth unemployment rate, such as overall economic stability and higher wages.

k. Merging population data

Go online and find a data set which contains the 2020 population for the countries of the world together with ISO codes.

- Download this data and merge it to the dataset you are working on in this case study using a left join. (A possible source: World Bank))
- Inspect the data and check whether the join worked well.

Answer

```
population_data <- read.csv("population_data_worldbankgroup.csv", header = TRUE, sep=";", skip = 4)

# only keep year 2020, renaming it for clarity
population_data <- population_data %>%
  select(Country.Code, X2020) %>%
  rename(population_2020 = X2020)

# merge population data with the main dataset, using a left join
data_case_study2 <- data %>%
  left_join(population_data, by = c("ISO3" = "Country.Code"))

# Check for missing values in the merged dataset
data_case_study2 %>%
  filter(is.na(population_2020))

## # A tibble: 1 x 11
##   country ISO3 continent subcontinent status expenditure youth_unempl_rate
##   <chr>   <chr> <chr>      <chr>          <ord>          <dbl>          <dbl>
## 1 Taiwan TWN Asia Eastern Asia H NA NA
## # i 4 more variables: net_migr_rate <dbl>, low_yu <fct>, high_nmr <fct>,
## #   population_2020 <dbl>

# check distribution of population to identify other possible issues
summary(data_case_study2$population_2020)

##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.      NAs
## 1.040e+04 8.048e+05 6.565e+06 3.630e+07 2.468e+07 1.411e+09      1

head(data_case_study2)

## # A tibble: 6 x 11
##   country ISO3 continent subcontinent status expenditure youth_unempl_rate
##   <chr>   <chr> <chr>      <chr>          <ord>          <dbl>          <dbl>
## 1 Afghanistan AFG Asia Southern As~ L 4.1 17.6
## 2 Albania ALB Europe Southern Eu~ UM 3.6 31.9
## 3 Algeria DZA Africa Northern Af~ LM NA 39.3
## 4 American Sa~ ASM Oceania Polynesia UM NA NA
## 5 Andorra AND Europe Southern Eu~ H 3.2 NA
## 6 Angola AGO Africa Sub-Saharan~ LM 3.4 39.4
## # i 4 more variables: net_migr_rate <dbl>, low_yu <fct>, high_nmr <fct>,
## #   population_2020 <dbl>
```

A single NA value suggests, that the left-join worked well. Further, all population values lie in a reasonable range.

1. Scatterplot of education expenditure and net migration rate in Europe

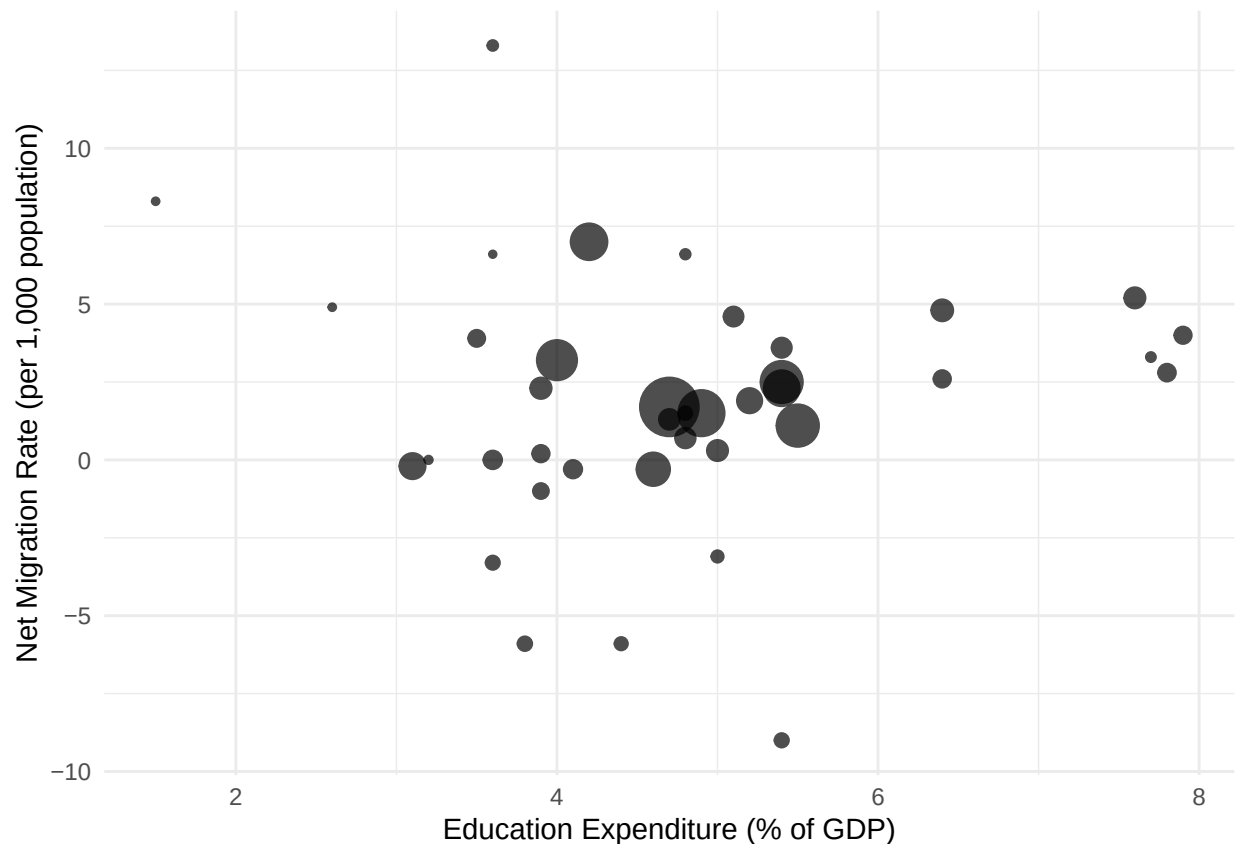
Make a scatterplot of education expenditure and net migration rate for the countries of Europe.

- Scale the size of the points according to each country's population.
- For better visibility, use a transparency of `alpha=0.7`.

- Remove the legend.
- Comment on the plot.

Answer

```
# Filter for European countries
europe_data <- data_case_study2 %>%
  filter(continent == "Europe", !is.na(expenditure), !is.na(net_migr_rate))
# Create the scatter plot
ggplot(europe_data, aes(x = expenditure, y = net_migr_rate)) +
  geom_point(aes(size = population_2020, alpha = 0.7)) + # relative size and adjusted alpha
  scale_size_continuous(range = c(1, 10)) + # control range of point sizes
  labs(
    x = "Education Expenditure (% of GDP)",
    y = "Net Migration Rate (per 1,000 population)"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none", # remove legend
  )
```



In the scatter plot, we notice, that countries with large populations are more centered. On the one hand, this means more consistent values for the migration rate, which makes sense as it is measured per 1000 population. On the other, we find relatively consistent amounts of money spend on education for high population countries (around 4-6%), while countries with less population show more variation.

From the scatter plot, we find a slight positive correlation between education expenditure and net migration rate. We could conclude, that boosting education budgets may help attract or retain people. However,

this trend is rather subtle in the data, and secondly, this might be heavily influenced by other factors and represent only a small picture of a far more complex setting.

m. Interactive plot

On the merged data set from Task k., using function `ggplotly` from package **plotly** re-create the scatterplot in Task l., but this time for all countries. Color the points according to their continent.

When hovering over the points the name of the country, the values for education expenditure, net migration rate, and population should be shown. (Hint: use the aesthetic `text = Country`. In `ggplotly` use the argument `tooltip = c("text", "x", "y", "size")`).

Hint: If you are creating a .pdf file, keep in mind that PDF output is static and does not support HTML widgets. To avoid build errors, you can wrap interactive plots in a conditional check or set the chunk option `eval=FALSE`. Alternatively, you can use the **webshot** package to automatically convert interactive elements into static images during the knit process.

Answer

```
library(plotly)

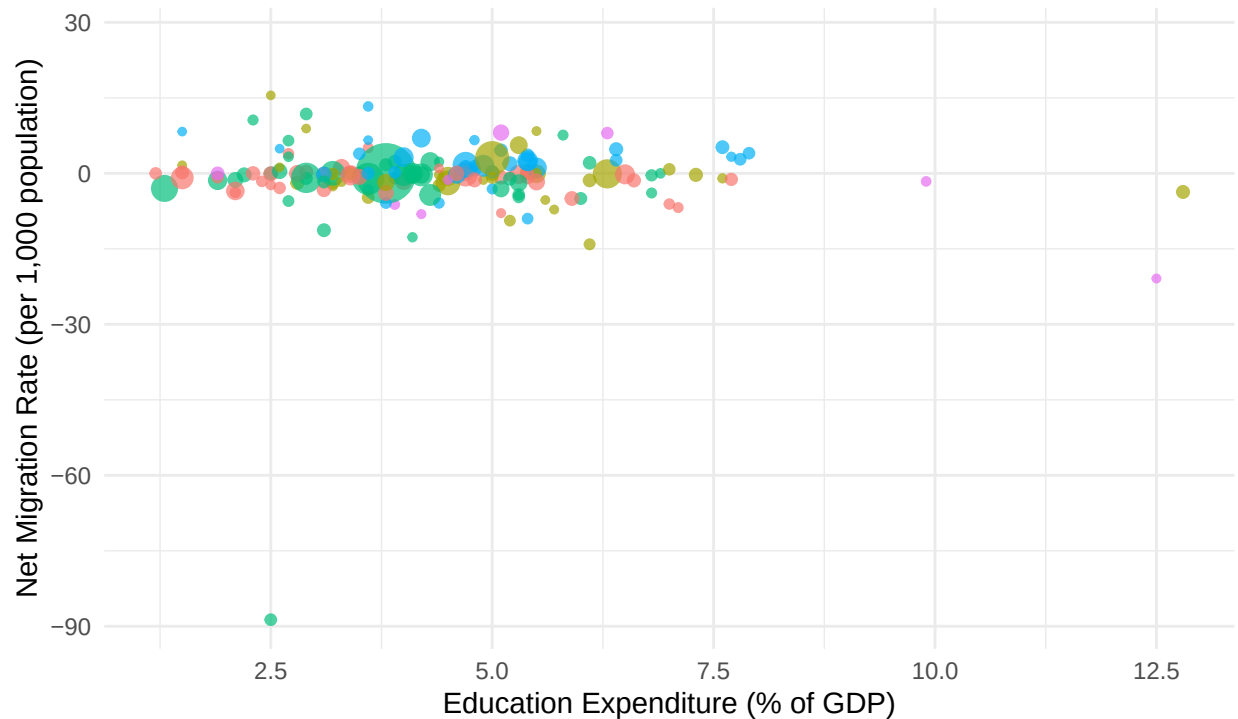
##
## Attache Paket: 'plotly'
## Das folgende Objekt ist maskiert 'package:ggplot2':
##
##   last_plot
## Das folgende Objekt ist maskiert 'package:stats':
##
##   filter
## Das folgende Objekt ist maskiert 'package:graphics':
##
##   layout

interactive_plot <- ggplot(data_case_study2, aes(x = expenditure, y = net_migr_rate, text = country)) +
  geom_point(aes(size = population_2020, color = continent), alpha = 0.7) + # color according to con
  scale_size_continuous(range = c(1, 10)) +
  labs(
    x = "Education Expenditure (% of GDP)",
    y = "Net Migration Rate (per 1,000 population)"
  ) +
  theme_minimal() +
  theme(
    legend.position = "top"
  )

if (knitr::is_html_output()) {
  ggplotly(interactive_plot, tooltip = c("text", "x", "y", "size")) %>%
    layout(
      xaxis = list(title = "Education Expenditure (% of GDP)"),
      yaxis = list(title = "Net Migration Rate (per 1,000 population)")
    )
} else {
  interactive_plot
}
```

```
## Warning: Removed 48 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

ation_2020 ● 5.0e+08 ● 1.0e+09 continent ● Africa ● Americas ● Asia ● Europe



n. Parallel coordinate plot

In **parallel coordinate plots** each observation or data point is depicted as a line traversing a series of parallel axes, corresponding to a specific variable or dimension. It is often used for identifying clusters in the data.

One can create such a plot using the **GGally** R package. You should create such a plot where you look at the three main variables in the data set: education expenditure, youth unemployment rate and net migration rate. Color the lines based on the income status. Briefly comment.

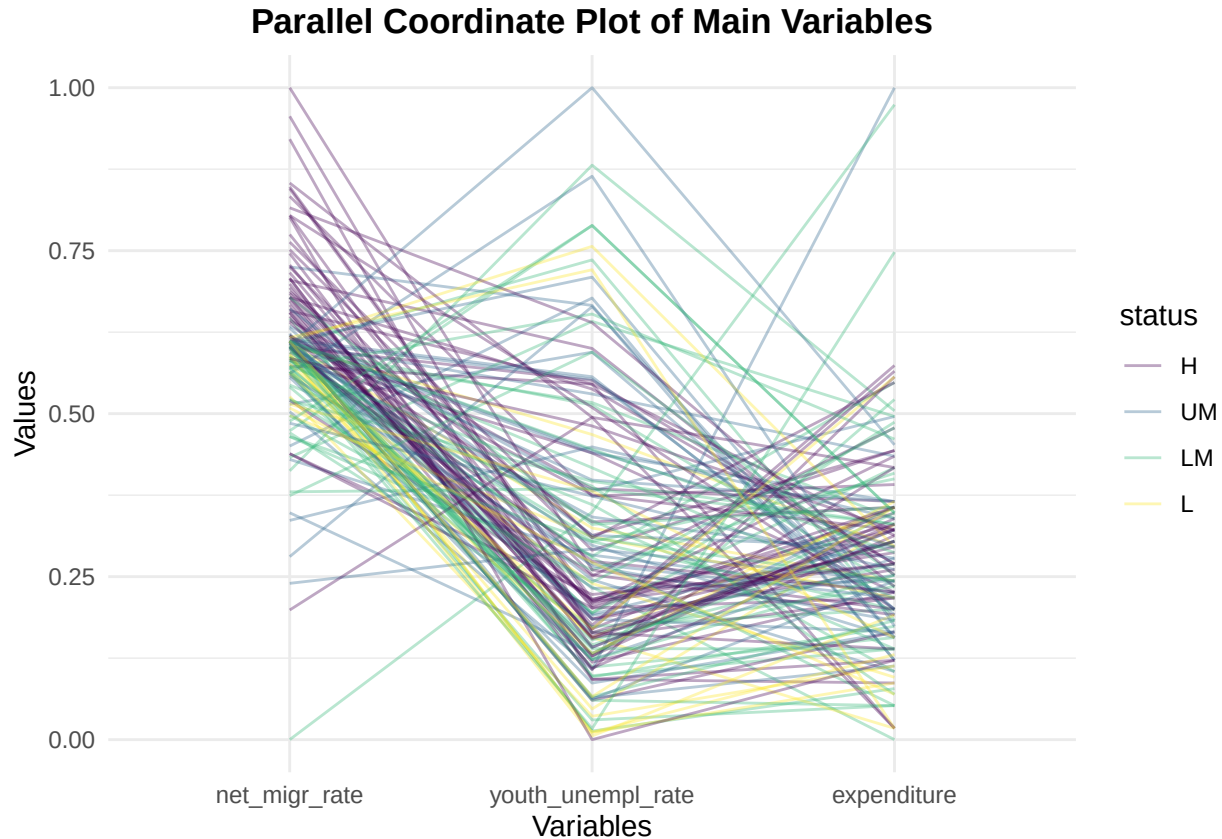
Answer

```
library(GGally)

# Create a parallel coordinate plot
ggparcoord(data_case_study2 %>% tidyr::drop_na(),
  columns = c("expenditure", "youth_unempl_rate", "net_migr_rate"),
  groupColumn = "status",
  scale = "uniminmax",
  alphaLines=0.35,
  order = "anyClass") +

labs(
  title = "Parallel Coordinate Plot of Main Variables",
  x = "Variables",
  y = "Values"
```

```
) +
theme_minimal() +
theme(
  plot.title = element_text(face = "bold", hjust = 0.5)
)
```



In the parallel coordinate plot, we colored the lines according to the income status of that country. However, as the lines of the different income groups heavily overlap, we can not identify any obvious income-related patterns or clusters in the data.

Many of the lines slope downward from a high `net_migration_rate` to low `youth_unempl_rate`, and upward from low `net_migration_rate` to high `youth_unempl_rate`. This hints a negative correlation between the two variables. However, as a parallel coordinate plot isn't ideal for checking correlation (e.g. because of the independently scaled variables), this trend should be checked statistically.

o. World map visualisation

Create a world map of the education expenditure per country. You can use the vignette <https://cran.r-project.org/web/packages/rworldmap/vignettes/rworldmap.pdf> to find how to do this in R. Alternatively, you can use other packages (such as `ggplot2`, `sf` and `rnaturalearthdata`) to create a map.

Answer

```
library(rworldmap)
```

```
## Lade nötiges Paket: sp
```

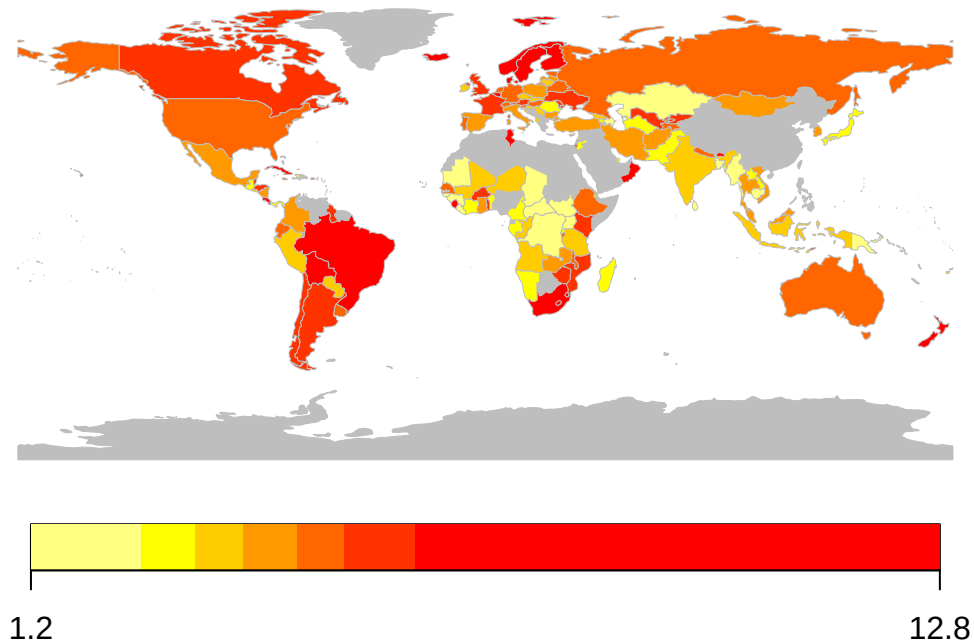
```
## ### Welcome to rworldmap ###
```

```
## For a short introduction type : vignette('rworldmap')
world_map <- joinCountryData2Map(data_case_study2, joinCode = "ISO3", nameJoinColumn = "ISO3")

## 215 codes from your data successfully matched countries in the map
## 1 codes from your data failed to match with a country code in the map
## 29 codes from the map weren't represented in your data

map <- mapCountryData(world_map,
  nameColumnToPlot = "expenditure",
  numCats = 7,
  mapTitle = "World Map of Education Expenditure (% of GDP)",
  colourPalette = "heat",
  missingCountryCol = "grey",
  addLegend = TRUE)
```

World Map of Education Expenditure (% of GDP)



The plot shows the education expenditure for each country on a world map, with small expenditures colored yellowish and high expenditures colored reddish.

We notice some expected trends: Many African, Middle-East and South-Asian countries have low education expenditures, while most countries in Europe and North America tend to invest more.

However, some trends are also surprising: Sub-Saharan Africa is mixed, with many south-eastern countries having high education expenditures, while many north-western countries tend to have lower. Further, most countries in South America, e.g. Brazil, have high education expenditures, although being known as developing countries, that still face many social and economic issues. One possible explanation could be that the education system is not well developed, but the government still spends a lot of money on it.